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Tracing Learning Over Time: A Model-Based Subsequent Memory Analysis

2.1 INTRODUCTION

Memories can be difficult to make and maintain. Even when we try to memorize something, we sometimes fail to encode it the first time or forget it later. This gives rise to a va-

riety of interesting dynamics as specific information that is available to us in our memory rises and falls over time.

The dynamic nature of our memory processes make it difficult to study the neural signals that are relevant for learning and memory. For example, a common approach in cognitive neuroscience is the “subsequent memory paradigm” that compares neuroimaging signals at the time of encoding between later remembered and forgotten items (Sanquist et al., 1980; Paller et al., 1987; Wagner, 1998; Davachi et al., 2003; Ranganath et al., 2003; Kim, 2011). While this approach has revealed several of the neural systems that may be involved in memory formation, the method is limited in its ability to account for the dynamics of learning and memory over time. In a typical memory experiment, a participant will study a number of items and then, after some delay, be asked to retrieve the memory, through a recognition or recall test. The behaviors on this test are informative about the memory status but an incorrect recall could come about for multiple reasons: the participant might have had a failure of retrieval, they might have encoded it but forgotten it during the delay or they might never have encoded it in the first place. Similarly, a correct response could be based on an actual retrieved memory or just a lucky guess.

Furthermore, the subsequent memory approach is difficult to generalize to cases when there are multiple presentations (so that the link between a specific pattern of neural activity and successful memory formation is less clear). While the effects of repeated practice and time since last study have been well known since Ebbinghaus (1913), there is no straightforward approach to applying a subsequent memory analysis to understanding the dynamic nature of our memory: how the probability of an item being remembered by a subject increases over several presentations and how time since study tends to reduce per-

formance.

The inherent challenge is that the dynamics of learning and memory typically cannot be directly measured. In a sequence of repeated study opportunities, how does one tell apart a study event where a memory was successfully formed from one where it wasn't when some things might have been easily learned on the first presentation, others over multiple trials and still others learned but then forgotten before the eventual test?

In other fields of cognitive neuroscience, sorting out the relationship between latent computations and brain activity has been advanced using *model-based* neuroimaging analyses. In this approach, quantitative cognitive models are fit to the behavior of the subject and then a time series of latent cognitive variables are extracted and compared to the time series of the neural signals (Daw, 2011). This approach has been particularly successful in the study of reinforcement learning and value-based decision making (O'Doherty et al., 2003; Daw et al., 2006, 2011) but has also recently been applied to several other fields where tracking latent dynamics is key such as perceptual decision making (van Maanen et al., 2011; Nunez et al., 2017) and category learning (Mack et al., 2013, 2016; Davis et al., 2014).

In the following work, we propose a new approach to tracking the latent dynamics of memories over time. Specifically, we build upon classic models of learning developed in the mathematical psychology literature and use them to structure a model-based analysis of an fMRI study of humans learning associations between words across several presentations. We propose that this new modeling approach may provide a way to generalize the subsequent memory approach to cases with multiple presentations and arbitrary delays between study and test and help identify signals in the brain that track the dynamics of learning and

memory over time.

2.1.1 PAST APPROACHES TO STUDYING THE DYNAMICS OF MEMORY OVER TIME

As mentioned earlier, the subsequent memory paradigm has most often been used to study neuroimaging signals related to mnemonic status (Sanquist et al., 1980; Paller et al., 1987; Wagner, 1998; Davachi et al., 2003; Ranganath et al., 2003; Kim, 2011) but has challenges being applied when there is variation in the amount of study or delay since last exposure.

Past studies involving multiple repeated study trials have compared signals across pairs of trials, typically looking at the similarity of multivariate signals across pairs (known as representational similarity analysis (Edelman et al., 1998; Kriegeskorte, 2008)) or difference in univariate patterns (targeting repetition suppression (Grill-Spector and Malach, 2001; Henson, 2003)) across pairs of study presentations (Xue et al., 2010, 2013; Ward et al., 2013; Hasinski and Sederberg, 2016; Lu et al., 2015; Feng et al., 2019). In some cases, these measures can be averaged across several presentations to derive a prediction for later memory (as in Xue et al., 2010) but this approach has rarely been justified on purely theoretical grounds. Overall, while these more model-free approaches have been very valuable for understanding how representations change, it is difficult to generalize beyond looking at pairs of repeated presentations.

Approaches based on model-based analyses have begun to be applied to memory studies, but while a few papers have applied principles inspired by models of recognition memory to analyzing fMRI data (Davis et al., 2014; LaRocque et al., 2013; Nosofsky et al., 2012), analyses that track memory dynamics in traditional tasks (like recognition and cued or free recall) on a trial-by-trial basis are still extremely rare (but see Kragel et al., 2015, for an analy-

sis of dynamics at retrieval in free recall). One recent relevant work studied cross-situational word learning (Berens et al., 2018), a paradigm where participants study words paired with stimuli (typically images) and later participants are given a cued recall test where they have to retrieve the word associated with a given stimulus. Unlike traditional memory paradigms though, several words and stimuli are presented at the same time during study so there is ambiguity about what the correct pair to learn is, adding an extra complication to the learning task. Berens et al. (2018) had participants complete several rounds of studying and then testing on the same set of nine word-stimuli pairs. They found signals in several regions that tracked the total amount of learning on a study trial (across all words) and showed that a cognitive model of learning predicted changes in the similarity across items in the left posterior hippocampus during the test trials over the course of several rounds. This corresponded well with a hypothesis that learning in the hippocampus often involves *pattern separation* from other studied material (LaRocque et al., 2013; Brunec et al., 2020).

These results are promising in that it suggests that there may be signals that track learning in related recall-based memory paradigms. However, if we want to directly track the learning that is happening on a particular study trial, the design of Berens et al. (2018) may be less than ideal. This is because of the well-known *testing effect* (Abott, 1909; Izawa, 1966; Tulving, 1967; Allen et al., 1969; Carrier and Pashler, 1992; Roediger and Karpicke, 2006b) where the act of attempting to retrieve can itself increase the probability of recall in the future. Therefore, to find signals that directly track learning on a trial-to-trial basis, we need a paradigm that only involves a final testing trial and a model that can make predictions on a trial-by-trial basis.

2.1.2 APPROACH

Before detailing our study design, we first give some intuition for our modeling approach. We assert that two types of information can inform the likely dynamics of memory: the average recall performance on an item in the population (its *memorability* (Bainbridge, 2019)) and the specific behaviors observed in a subject, not limited to a memory test alone.

It is well-known that items vary in their intrinsic memorability — their probability of being later recognized or recollected across people — across a variety of memory tasks (Glaze, 1928; Baldwin and Runkle, 1967; Rubin, 1985; Isola et al., 2011; Bainbridge, 2019). If we assume a monotonic shape of a learning curve, this memorability dictates a significant amount of the dynamics of learning. For instance, if two items are both successfully recalled, the item that was easier to learn is likely to have been learned earlier. Similarly, if two items were both forgotten, the easier to learn item is more likely to have been learned and then forgotten while the more difficult one might never have been encoded at all. In fact, while memorability is often studied as a unitary construct (Bainbridge, 2019), memorability is itself dependent on an item’s probability of being encoded on a single study opportunity (which we call *learnability*) and its probability of being forgotten (which we call *forgettability*). While there is very little past work on this (for one of the few examples, see Woodworth, 1914), these two variables do not have to be perfectly correlated. In order to investigate this and make more precise predictions about memory dynamics, we separate them in our modeling approach. Overall, this intuition that item-level effects are important for tracking learning suggests that accounting for these variables can give us a clearer picture of memory dynamics over a straightforward application of a subsequent memory approach which would consider each trial to be equally related to later recall and does not typically

consider variation in memorability across studied items.

We can combine information from learnability and forgettability with the intuition from the subsequent memory paradigm that later recall might suggest different memory states at the time of the initial exposure. If two people studied the same item but only one of them later remembered it, the probability of having encoded it after each study opportunity should be higher for the person who did later recall it. We can generalize this idea to additional behavioral measures such as judgments of learning (JOLs) (a judgement of how likely a subject is to later remember an item), which are known to predict later memory (Arbuckle and Cuddy, 1969; Lovelace, 1984; Nelson and Dunlosky, 1991). If one participant has more confidence in their learning, they are similarly likely to have a higher probability of having encoded the item. We are able to capture all of these ideas quantitatively with simple measurement models that we describe in more detail below.

STUDY DESIGN

In order to identify fMRI signals that track learning over several trials, we use a classic paired associates learning task where participants repeatedly study pairs of words, one pair at a time. Participants completed the study session in the fMRI scanner and then returned to the lab three days later for a cued recall test.

In general, encoding quality for a particular item in paired associates tasks will depend on the particular stimuli (Xie et al., 2020, i.e. memorability,) as well as the order in which they are presented (Davis et al., 2008). Because our primary interest in this study is to track the mnemonic status of particular items rather than understanding the effects of context or other experimental manipulations, we take the unusual step of presenting the same items

in the same order for every participant in our task. At the expense of generalizability, this allows us to estimate precise learning trajectories for each item in its particular context. * Having more precise estimates will increase our power to find signals (Wilson and Niv, 2015, and see Chapter 3 of this thesis).

Another key feature of our design was to collect a large dataset of subjects who did not perform either part of the study in the scanner. All of the scanner subjects were collected at the same delay between study and test, in order to optimize our ability to predict memory based on the fMRI signals. This prediction is not the focus here and will be reported in other papers (Chapter 1). However, one aspect of memory that we wanted to capture with our model was the possibility that participants encoded but then forgot an item during the delay. In order to fit a forgetting curve, we needed to have participants with significant variation in delays. Therefore, we collected data from a large number of subjects outside of the scanner at varying delays in order to be able to fit this part of the model (see Chapter 3 for a detailed examination of the advantages of this approach).

MEASUREMENT MODEL

We construct a cognitive measurement model to account for the dynamics of memory for each item studied in our experiment based on Markov models of paired associates developed in the 1960s (Bower, 1961; Atkinson and Crothers, 1964; Calfee and Atkinson, 1965), which are themselves based on Estes' Stimulus Sampling Theory (Estes, 1950; Estes and Burke, 1953; Estes, 1955). The key idea in these models is that learning is an all-or-

*Although the models we apply do not explicitly model inter-item interactions, maintaining a fixed sequence across participants ensures that some of these inter-item effects will be captured in the model parameters we estimate because, for instance, the measured difficulty of a word is always assessed with respect to the other list items.

none phenomenon such that any pair of items to be learned is in one of only two *mnemonic states* at any given time, *(U)nlearned* (denoted s_U) and *(L)earned* (denoted s_L). Prior to any learning, all items have a very high probability of being in the s_U state. When a study trial occurs, item w has a fixed probability, l_w of learning, i.e. transitioning into the s_L state. According to these models, when a test trial occurs, items in the s_L state will receive a correct response while items in the s_U state will only be correct due to random guessing.

Perhaps somewhat surprisingly, early versions of these models discussed above did not incorporate the dynamics of forgetting. This is perhaps because many of the experiments at the time used relatively short study-test delays such that forgetting was minimal. [Atkinson and Crothers \(1964\)](#) altered the model by including the possibility that, for all items not studied on a particular study trial, items that were in higher mnemonic states could transition to lower ones with probability f_w . [Calfie and Atkinson \(1965\)](#) provided further evidence that this improved the fit of the model.

While very simple, there are several reasons this two-state Markov model provides a good measurement model of memory dynamics for our purposes. It easily allows for modeling the learnability, l_w , and forgettability, f_w , of individual word pairs using a minimal number of parameters. In addition, because the model can be written as a Markov process ([Bower, 1961](#)), we can use established techniques ([Baum and Eagon, 1967](#); [Baum et al., 1970](#)) to derive predictions for the fMRI time series that are conditioned both on the parameters relating to the word pair being studied as well as each individuals data including recall tests and judgments of learning. As we mentioned above, these two functions are key to our ability to capture the latent memory dynamics.

In addition, this model provides a relatively good account of the data. At the time it

was developed, the [Bower \(1961\)](#) model accounted for a surprising amount of the dynamics of learning in paired associates, including how many items were remembered after a certain number of trials, the probability that no item would be incorrectly recalled after a certain number of study trials, and compared favorably to alternative models such as the linear model based on [Bush and Mosteller \(1953\)](#) ([Suppes and Ginsberg, 1963](#); [Crothers, 1962](#); [Bower, 1962](#); [Estes, 1961](#)). Because of the restricted nature of our task, more complex models cannot provide a significantly better explanation for our data. For instance, while [Atkinson and Crothers \(1964\)](#) and ([Calfee and Atkinson, 1965](#)) suggest that more states are needed to capture memory dynamics in paired associates, these differences can only be identified with more test trials (see [Greeno, 1968](#); [van De Sande, 2013](#); [Feng, 2017](#), for more on these issues). A linear model like the one from [Bush and Mosteller \(1953\)](#) is in fact equivalent to this one in terms of predictions about accuracy on a single test trial ([Restle, 1965](#)).[†] Similarly, because of the fixed order and timing of the stimuli and the single test trial, predictions from more modern models such as [Raaijmakers \(2003\)](#) or [Pavlik and Anderson \(2005\)](#) will either be very similar or will require too many parameters to be estimated from the data. Because the model is formally or nearly equivalent to many other cognitive process models, we refer to it as a cognitive measurement model, used for measuring the latent dynamics of memory rather than making a claim about the representations used in memory processes. Finally, despite largely falling out of favor in psychology in the 1970s for partly sociological reasons (see [Bower, 1994](#)), a model that is equivalent to [Bower](#)

[†]Despite this equivalency in forward prediction, in this case, we prefer the discrete-state model because of our ability to easily infer likely past states given observed recall using properties of Markov models. For continuous state models, this is not yet possible with arbitrary observations (like a binary recall variable). In general, this would require approximate methods like particle filters but see ([Linderman et al., 2015](#)) for a promising future direction.

(1961), known as Bayesian Knowledge Tracing (Corbett and Anderson, 1995), has been successfully used for making predictions about learning and choosing what to study in automated tutoring systems for over 25 years, providing further evidence that the model can capture the broad strokes of learning dynamics.

MODEL-BASED PREDICTORS

In order to use the model for structuring an fMRI analysis, we need to derive trial-by-trial predictors from our model. We choose two variables from the model that are likely to be reflected by brain regions related to memory status.

$P(s_{Lpost})$ The first is simply the probability of being in the *Learned* state (s_L) after a study trial (i.e., *post*). This directly depends on the *learnability* of a given word-pair relative to the other pairs being studied as well as how many study repetitions have already been completed. In general, more learnable words will be more likely to be in the Learned state early in the experiment while others will catch up later. In addition, it will also depend on a particular subject's measures of recall and confidence as these will adjust the overall probability that a pair was learned by the end of the experiment.

$P(switch)$ This is the probability that a switch from Unlearned (s_U) to Learned (s_L) occurs on a particular study trial. While it may seem that $P(switch)$ should be perfectly correlated with $P(s_{Lpost})$, this is not entirely the case. In general, a switch will be more likely to occur early in the experiment (when every word pair is unlearned) but more so for more learnable words. By the last study trial, a switch actually may be

more likely for less learnable word pairs whereas more learnable words will always be more likely to be in the s_L state.

These two cognitive variables would both be consistent with many of the observed subsequent memory effects we discussed above as, on a single study trial, a trial that is later remembered would both involve a state switch and also result in being in the learned state after the trial. However, because they become uncorrelated over many study trials in the model, this study provides us with a unique opportunity to dissociate these two distinct computations which could be occurring in different regions.

In particular, one interesting interpretation of the $P(s_{L_{post}})$ variable would be in line with recent hypotheses that the fidelity of reinstatement of previous experiences with an item are a key mechanism for the effects of practice (Thios and D’Agostino, 1976; Benjamin and Tullis, 2010). A higher $P(s_{L_{post}})$ would then suggest a higher likelihood that there are previous experiences to reinstate and that the memory can be strengthened. This hypothesis has recently been argued to be supported by evidence that greater pattern similarity across trials in many brain regions supports memory (Xue et al., 2010, 2013; Ward et al., 2013). In addition, the parietal lobe is thought to be key to reinstatement and univariate activity in the parietal lobe has been shown to increase for repeated stimuli (Gilmore et al., 2019a), for vivid recollection (Lee et al., 2019; Daselaar, 2009) and also promote pattern similarity in other regions as well as future memory (Xue et al., 2013).

2.2 METHODS

2.2.1 PARTICIPANTS

Sixty-nine fMRI participants were recruited using electronic advertisements hosted by New York University and accessible to the broader community. We obtained written informed consent prior to conducting the study, in compliance with the New York University Institutional Review Board. All participants self reported to be between 18-35 years old, had normal or corrected-to-normal vision, spoke English fluently and did not speak Lithuanian or a related language.

Eleven participants' data were excluded from analysis based on MRI data issues: two participants' MRI sessions were conducted with incorrect scanning parameters; six participants were excluded based on excessive motion and other image quality issues; three participants requested early termination of the experiment due to discomfort in the scanner (and did not complete the recall test). An additional five participants were excluded from analyses based on lack of variation in behavioral measures that precluded inclusion in the two-stage general linear model analyses (within participant regression measures carried to group level analysis). One of these participants recorded 100% correct in the recall test and four participants gave Judgments of Learning (JOLs) of 100 to all word pairs (overall or within one condition). The lack of variation in these scores meant that no within-participant statistics could be computed for these measures' relationship to brain data and these participants could not be included in the group level random effects models. For consistency in the number and identity of participants included in various models we therefore opted to exclude these participants from subsequent neuroimaging analysis. With

these exclusions, this resulted in 53 total fMRI subjects included in the data set.

As mentioned above, in addition to the subjects who completed the task in the scanner, we also collected data from 148 behavioral participants who completed the same task in the lab but not in the fMRI scanner. These subjects were recruited using the same approach as for the fMRI subjects and all of these participants also self reported to be between 18-35 years old, had normal or corrected-to-normal vision, spoke English fluently and did not speak Lithuanian or a related language.

2.2.2 LANGUAGE LEARNING TASK

The task is outlined in Figure 2.1. On Day 1, fMRI participants ($N=69$) were scanned while performing a paired associates encoding task where they learned to associate English words with their Lithuanian translation. Starting with a normed set of Lithuanian-English words, we selected 45 translation pairs (Grimaldi et al., 2010). During study, participants saw the translation pairs presented one at a time for 4 seconds each with a variable duration inter-trial interval (4s-16s). Words were presented on a computer screen with the Lithuanian word at the top of the screen and the English translation underneath. During the inter-trial interval, in order to prevent rehearsal, participants made judgments about whether each of a series of numbers were odd or even.

Each word pair was presented five times and no pair was presented for the n th repetition until all words had $n - 1$ presentations. Importantly, and in contrast to many psychology studies on the subsequent memory effect, all participants see the same sequence of study items. Immediately following the study session participants gave judgments of learning (JOLs, (Nelson and Dunlosky, 1991)): for each pair participants were presented with the

Lithuanian and English word and used the computer mouse to indicate on a scale of 0-100 how likely they were to remember the association at the delay they were assigned for Day 2 (24, 72 or 168 hours). Participants had up to twelve seconds to respond and the response was coded as missing if this deadline wasn't met (in practice all participants gave all JOL ratings within the allotted time).

On Day 2, participants in the fMRI study returned to the lab approximately 72 hours after the initial study session to complete a recall test outside the fMRI scanner. During the recall test, participants saw a Lithuanian word presented on the screen and had to type the associated English word. Separate analyses of data from these neuroimaging subjects are also reported in Chapter 1.

As mentioned above, we also conducted a large behavioral experiment outside of the MRI scanner. These participants performed the exact same task as the fMRI participants, including the precise timing of the word-pair study trials on Day 1. Each participant ($N=148$) was tested at one of three study-test delays, 24 hours, 72 hours and 169 hours. Among the behavioral participants the group N s were 49, 50, and 49 in the 24, 72, and 168 hour study-test delay groups, respectively.

All of the experiments in both the fMRI and behavioral studies were coded using PsychoPy (Peirce, 2007) in python 2.7 (Rossum et al., 2010). All further analyses were conducted using python 3.7 and used the packages pandas (McKinney, 2010) NumPy (van der Walt et al., 2011) and SciPy (SciPy 1.0 Contributors et al., 2020). All plots of the behavioral data and model fits were made using seaborn (Waskom et al., 2020).

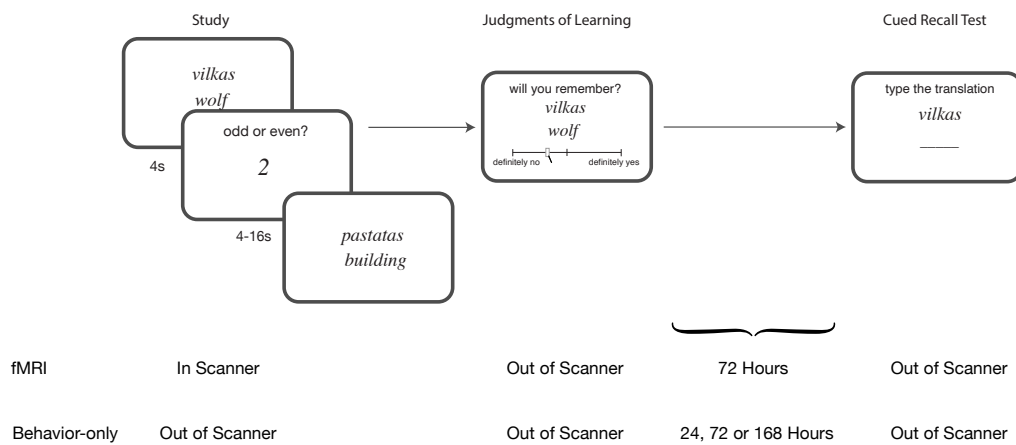


Figure 2.1: Experimental Design. On Day 1, subjects first completed a paired associates learning task in which an English word and a Lithuanian word were simultaneously presented on the screen. In between trials, subjects completed an odd-even judgment about a series of numbers in order to prevent rehearsal. For fMRI subjects this part of the task was completed in the scanner, for behavior-only subjects, it was completed outside the scanner. After the learning session, subjects made a judgment of learning about each of the paired associates. For all subjects this and Day 2 were completed outside the scanner. On Day 2, subjects completed a cued recall test in which they were presented with a Lithuanian word and typed in the corresponding English word. Day 2 was 72 hours after Day 1 for fMRI subjects and 24, 72 or 168 hours after for behavior-only subjects.

2.2.3 BEHAVIORAL PREPROCESSING

For scoring the recall test, a trial was coded as correct if participants typed the correct English word. In order to check correctness, all entries were converted to lowercase and spellchecked using the Enchant dictionary accessed through the python package pyenchant pyenchant[‡], complemented by a final check by hand. All other responses were incorrect. For inclusion in the model, Judgments of Learning were normalized to be between 0 and 1 by dividing by 100. Finally, throughout, a word-pair's *memorability* is always defined as the average recall performance across behavioral subject's in the 72-hour delay condition.

2.2.4 fMRI METHODS

MRI DATA ACQUISITION

MRI data were acquired on a 3 Tesla Siemens Prisma scanner. Anatomical images were collected using a T1-weighted MPRAGE high-resolution sequence (0.8mm isotropic voxels). Five runs of whole brain BOLD functional data were acquired using an EPI sequence (2.5mm isotropic voxels; TR=1s; TE=35ms; Multiband factor=4; Phase Encoding: Anterior-Posterior). An additional pair of non-accelerated EPI scans with opposing forward (anterior-posterior) and reverse (posterior-anterior) phase encoding relative to the primary functional data were collected for distortion correction during preprocessing (2.5mm isotropic voxels; TR=4.496; TE=45.6). The slice position and orientation for these scans matched those of the BOLD data collected during the behavioral task.

[‡]<https://github.com/AbiWord/enchant>, <https://github.com/pyenchant/pyenchant>

PREPROCESSING

Results included in this manuscript come from preprocessing performed using fMRIPrep 1.2.6-1 (Esteban et al., 2019), which is based on Nipype 1.1.7 (Gorgolewski et al., 2011).

ANATOMICAL DATA PREPROCESSING The T1w anatomical scans were corrected for intensity non-uniformity (INU) using N4BiasFieldCorrection (ANTs 2.2.0, Tustison et al., 2010). A T1w-reference map was computed (after INU-correction) using `mri_robust_template` (FreeSurfer 6.0.1, Reuter et al., 2010). The T1w-reference was then skull-stripped using `antsBrainExtraction.sh` (ANTs 2.2.0), using OASIS as target template. Brain surfaces were reconstructed using `recon-all` (FreeSurfer 6.0.1, Dale et al., 1999), and the brain mask estimated previously was refined with a custom variation of the method to reconcile ANTs-derived and FreeSurfer-derived segmentations of the cortical gray-matter of Mindboggle (Klein et al., 2017). Spatial normalization to the ICBM 152 Nonlinear Asymmetrical template version 2009c (Fonov et al., 2011) was performed through nonlinear registration with `antsRegistration` (ANTs 2.2.0, Avants et al., 2011), using brain-extracted versions of both T1w volume and template. Brain tissue segmentation of cerebrospinal fluid (CSF), white-matter (WM) and gray-matter (GM) was performed on the brain-extracted T1w using `fast` (FSL 5.0.9 Zhang et al., 2001).

FUNCTIONAL DATA PREPROCESSING For each of the five BOLD runs collected per subject (across all tasks and sessions), the following preprocessing was performed. First, a reference volume and its skull-stripped version were generated using a custom methodology of *fMRIPrep*. A deformation field to correct for susceptibility distortions was estimated

based on two echo-planar imaging (EPI) references with opposing phase-encoding directions, using 3dQwarp (AFNI 20160207, [Cox, 1997](#)). Based on the estimated susceptibility distortion, an unwarped BOLD reference was calculated for a more accurate co-registration with the anatomical reference.

The BOLD reference was then co-registered to the T1w reference using `bbregister` (FreeSurfer) which implements boundary-based registration ([Greve and Fischl, 2009](#)). Co-registration was configured with nine degrees of freedom to account for distortions remaining in the BOLD reference. Head-motion parameters with respect to the BOLD reference (transformation matrices, and six corresponding rotation and translation parameters) are estimated before any spatiotemporal filtering using `mcflirt` (FSL 5.0.9, [Jenkinson et al., 2002](#)). The BOLD time-series were resampled onto their original, native space by applying a single, composite transform to correct for head-motion and susceptibility distortions. These resampled BOLD time-series will be referred to as preprocessed BOLD in original space, or just preprocessed BOLD. The BOLD time-series were resampled to MNI152NLin2009cAsym standard space, generating a preprocessed BOLD run in MNI152NLin2009cAsym space.

Several confounding time-series were calculated based on the preprocessed BOLD: framewise displacement (FD), DVARS and three region-wise global signals. FD and DVARS are calculated for each functional run, both using their implementations in Nipype (following the definitions by [Power et al., 2014](#)). The three global signals are extracted within the CSF, the WM, and the whole-brain masks. Additionally, a set of physiological regressors were extracted to allow for component-based noise correction (CompCor, [Behzadi et al., 2007](#)). Principal components are estimated after high-pass filtering the preprocessed BOLD

time-series (using a discrete cosine filter with 128s cut-off) for the two CompCor variants: temporal (tCompCor) and anatomical (aCompCor). Six tCompCor components are then calculated from the top 5% variable voxels within a mask covering the subcortical regions. This subcortical mask is obtained by heavily eroding the brain mask, which ensures it does not include cortical GM regions. For aCompCor, six components are calculated within the intersection of the aforementioned mask and the union of CSF and WM masks calculated in T1w space, after their projection to the native space of each functional run (using the inverse BOLD-to-T1w transformation). The head-motion estimates calculated in the correction step were also placed within the corresponding confounds file. All resamplings can be performed with a single interpolation step by composing all the pertinent transformations (i.e. head-motion transform matrices, susceptibility distortion correction, and co-registrations to anatomical and template spaces). Gridded (volumetric) resamplings were performed using `antsApplyTransforms` (ANTs), configured with Lanczos interpolation to minimize the smoothing effects of other kernels (Lanczos, 1964). Non-gridded (surface) resamplings were performed using `mri_vol2surf` (FreeSurfer). Preprocessed BOLD data were smoothed to 6mm FWHM using AFNI `3dBlurToFWHM`[§].

2.2.5 MARKOV MODEL OF MEMORY

The following section describes the basic mathematical structure of our model. The core of our approach assumes a probabilistic representation of the latent mnemonic status (e.g., *remembered* versus *forgotten*) of each item on the to-be-remembered list. The model we use combines the simple two-state version of a Markov model of paired associates from Bower

[§]https://afni.nimh.nih.gov/pub/dist/doc/program_help/3dBlurToFWHM.html

(1961) (also known as the *one-element pattern model*) with the forgetting transitions proposed by Atkinson and Crothers (1964) in the context of slightly richer versions of the model.

Where our use of the model differs from past approaches is that we propose a mapping between these latent mnemonic states and other behavioral data like judgments of learning to inform inference about latent states. In addition, our use of the forward-backward algorithm (Baum and Eagon, 1967; Baum et al., 1970) to derive individualized predictions for the neuroimaging data collected during the study trials is novel relative to past applications where the main goal was to make forward predictions about future performance. Finally, unlike Calfee and Atkinson (1965) and Atkinson and Crothers (1964), we record the exact timing of each presentation so we are able to fit the forgetting curve in terms of seconds rather than stimulus presentations.

Each memory trace for a particular word-pair, w , (i.e., memory for the association between two words) in subject i can be represented as a non-homogenous, censored Markov model with the following properties:

STATES

Each trace can be in one of a number of discrete mnemonic **states**, $\mathcal{S}$, where $\mathcal{S} = \{s_U, s_L\}$ model with states corresponding to *unlearned* and *learned*. (see Figure 2.2). Following (Bower, 1961), s_L represents memories that are accessible and have recall probabilities equal to 1, (i.e., $\Pr[\text{recall} = \text{correct} | s_L] = 1$), The current state of word pair w at time t in subject i will be denoted $q_t^{j,w}$.

PRIORS

A **prior**, $\pi_{t=0}$, that captures our initial belief of the memory state of all items. The prior state for a particular memory for word-pair w , in subject i can be written as $\pi_{t=0}^{w,i} = \Pr[q_{t=0}^{w,i} = s]$ for $s \in \{s_U, s_L\}$. With unfamiliar learning materials we assume that the initial memory status is heavily biased towards the unknown state (i.e., $\pi_{t=0}^{i,s_U}$ is much higher $\pi_{t=0}^{i,s_L}$) and also that these probabilities are the same across subjects and word pairs.

TRANSITIONS

A set of **transition probabilities**, $\mathcal{A}$, which determine the likelihood that a memory will move between the different states at each time point. In prototypical Markov models, the transition probabilities are stationary and the same transitions are applied at each time step. In this model, there are different sets of transition probabilities which are applied at a given time step depend on the type of external “event”, $e_t^{w,i}$, that occurs (e.g., a study trial versus a time step between trials; Figure 2.2). For a word pair w in subject i , the transition probability of moving from state s to s' after an event of type g will be denoted $a_t^{w,i,g,s \rightarrow s'} = \Pr[q_t^{w,i} = s' | e_t^{w,i} = g, q_{t-1}^{w,i} = s]$ where g indicates the specific event type for word pair w and subject i at time t .

Event types depend on the particular experiment design but here include “study trial” (*study*), “study with JOL trial” (*study+JOL*), “timestep in which memory decays” (*decay*), and “test trial” (*test*). Generally, during *study* or *study+JOL* events we assume that items tend to transition from a more poorly learned state to a more fully learned state. The probability of transitioning to the learned state on a study trial is represented by parameter l_w (see Figure 2.2). During *decay*, items in (s_L) have a probability of transitioning to the *un-*

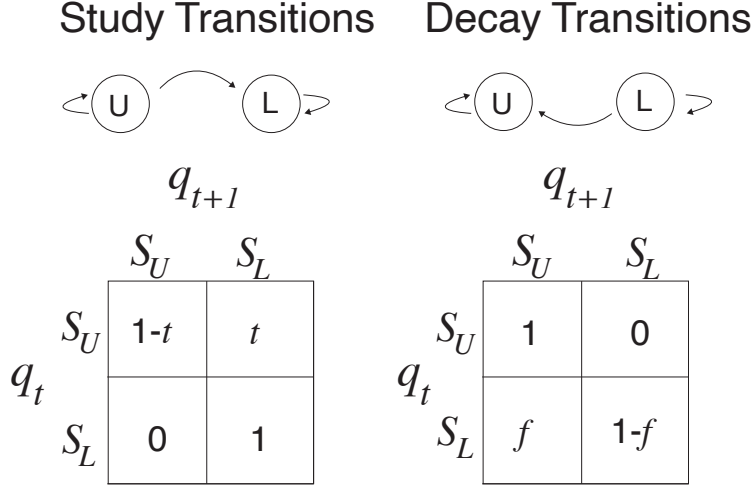


Figure 2.2: The matrix of transition probabilities for either study or decay events. The letters within each matrix reflect the transition parameters which are estimated to data. The state labels s_U are *Unlearned*), s_L are *Learned*)

known state with probability f_w . Decay events are necessary to account for the patterns of forgetting across the study-test delay intervals shown in Figure 2.3. We assume test trials have no effect on transitions as they appear at the end of the task.

We define an experiment **protocol** for subject i , $\mathbf{E}_i$, as a $W \times T$ matrix where W is the number of items being studied and T is the total number of micro-time steps modeled in the experiment. Each entry of the matrix, $e_i^{t,w}$, codes which of a discrete set of event types occurred on a time step as described above. While in the experiment, *study* trials lasted for 4 seconds, only the entry in the protocol that corresponded to the beginning of the study trial was coded as a *study* event. All others after were coded as *delay*. While the scale of forgetting over 4 seconds is negligible, future versions of the model could improve on this assumption. Similar assumptions were implemented for *study+JOL* and *test* events. The pro-

protocol captures the dependencies between event sequences that influence different memory traces. For example, if word-pair w is studied on a given trial, then all the other items on the list might undergo a memory decay event during the same time step. This way, the protocol enforces the implicit tradeoffs of studying one item over others at a particular point in time. In our experiment, all subjects saw items in the same order but there was slight variation in how long the breaks were between blocks of study trials as well as in the delay between the last study trial, the JOL trial and the final test. This protocol allows the model to capture the different predictions for each subject depending on their specific timing.

OBSERVABLE SIGNALS

The mapping between the latent states $\mathcal{S}$ and behavior is made through a set of **observation distributions**, $\mathcal{B}$, which define the probabilities that, on event type g at time t in subject i , an observable random variable of data type d , $\mathbf{o}_{t,i}^{g,d}$ takes on a value $v_k^{g,d}$ from an (potentially infinite) alphabet $\mathcal{V}^{g,d}$. For each memory trace i , we can write the probability of its associated observables as $b_t^{i,g,s,d}(v_k^{g,d}) = \Pr[o_t^{g,d} = v_k^{g,d} | e_t^i = g, q_t^i = s]$. Observation distributions in effect define the full generative model that links both behavior and neural information to underlying knowledge states. Here we consider two types of observations: recall tests (*recall*) and JOLs (*JOL*). However, this approach could easily incorporate many other measures such as response times or pupil dilation [‡].

Recall Test. At the end of the experiment, the last entry in the protocol is a recall test where the subject is asked to recall a studied item from memory. The response given on these trials is treated as an observation associated with this particular type of event. Specifi-

[‡]Of course, the fMRI signal itself could play this role but the high dimensionality and uncertainty about which aspects of the signal are relevant makes this difficult. For more on this point, see the discussion.

cally, the alphabet is $v^{test,recall} \in \{correct, incorrect\}$ and $v^{g,recall} \in \emptyset$ for $g \neq test$, reflecting the absence of any recall response on non-test events. The distribution of test question answers about memory trace i from state s at time t , is then $b_t^{i,test,s,recall}(correct) = p_{recall,s}$ and $b_t^{i,test,s,recall}(incorrect) = 1 - p_{recall,s}$ where $p_{recall,s}$ is defined (or fitted) for each memory state. For other trial types, i.e. $g \neq test$, $b_t^{i,g,s,recall}(\emptyset) = 1$. So the posterior probability of being in a state s on those events is driven by the state transitions or other observations.

Judgments of Learning. JOL responses were only given on the last study trial (a *study+JOL* event). JOLs normalized to be between 0 and 1 were included in the model as an observation with $v^{study+JOL,JOL} \in [0, 1]$ and null for other trial types. We model the distribution of JOLs given the state as a truncated Gaussian distribution in the range 0 to 1, i.e. $b_t^{i,study+JOL,s,JOL} = TN(\mu_{JOL,s}, \sigma_{JOL,s}, 0, 1)$ with $\mu_{JOL,s}$ and $\sigma_{JOL,s}$ defined independently for each state s .

INFERENCE

The full model is specified by a protocol, E , a set of priors over the states, $\pi_{t=0}$, a set of transition probabilities, A , and a set of observation distributions associated with each state-event pair, B . For fitting the model using the forward algorithm (Baum and Eagon, 1967), we need a conditional probability of being in a state s at time t . Using Bayes' rule, the probability that a memory trace for word pair w in subject i on trial t is in state $s' \in S$ given the parameters and past observations is:

$$\pi_t^{w,i,s'} = \frac{b_t^{w,i,g,s',d} a_t^{w,i,g,s \rightarrow s'} \pi_{t-1}^{w,i,s}}{\sum_j b_t^{w,i,g,s_j,d} a_t^{w,i,g,s \rightarrow s_j} \pi_{t-1}^{w,i,s}} \quad (2.1)$$

Combining this with the observation distributions described above, we can get an un-

normalized posterior for a particular parameter combination given the data that allows us to fit the model.

Model parameterization. Partially due to identifiability concerns (van De Sande, 2013; Feng, 2017), some parameters were fixed to semantically coherent values (Doroudi and Brunskill, 2017), while others were estimated from the data.

For all words we fixed the initial state priors, $\pi_{t=0}$, as $[\cdot.99, \cdot.01]$ for s_U, s_L in the two-state model. This was motivated by the fact that none of the learners in our dataset had prior experience with Lithuanian. Following Bower (1961), we also fixed the probabilities of giving the correct test response, $\mathbf{p}_{\text{recall}}$ as $[\frac{1}{|\mathcal{W}|}, 1] = [\frac{1}{45}, 1]$ for latent memory states s_U and s_L . This reflects the assumption that it is very unlikely that one would guess the correct answer in a cued recall test without any memory and that we consider memory for individual English words from study but not for the correct association to be not truly learned.

Fitted parameters include those determining the transition probabilities and observation distributions within each model. Transition probabilities are fit for each word pair w (summarized in Figure 2.2). These are the l_w and f_w parameters controlling memory strengthening and decay, respectively.

We fit our models using Bayesian inference which requires a prior on individual model parameters. We used a uniform prior on the interval $[0, 1]$ for all l_w and f_w parameters. Future work would likely benefit from a hierarchical approach to inferring these parameters (Rouder and Lu, 2005; Ahn et al., 2011) but in practice the geometry of the posteriors in our problem made this difficult to implement computationally (for more on the challenges of sampling for hierarchical models, see Betancourt and Girolami, 2015).

When fitting models that incorporated JOLs we also estimated the means and variance

parameters for the Gaussian (truncated for JOLs) observation likelihood from each latent state. For the JOL distributions, each μ_{JOL_s} was drawn from a $\text{Normal}(.5, .5)$ and each σ_{JOL_s} was drawn from $\text{Exponential}(.5)$.

Model estimation. We used MCMC sampling via the dynamic Hamiltonian Monte Carlo algorithm (Betancourt, 2018), as implemented in Stan (Carpenter et al., 2017) and accessed through `pystan`, to estimate the posterior over the parameters (4 chains of 300 iterations; 150 per chain discarded as burnin; 600 total samples per parameter). To ensure convergence, we checked that our sampler did not observe any divergences (Betancourt, 2018) and that estimates had folded, split, rank-normalized $\hat{R}$ values (a measure of whether the sampling chains are converging to similar estimates) < 1.05 (Vehtari et al., 2020; Gelman and Rubin, 1992).

Model comparison. To check whether the use of the judgment of learning data improved the model’s ability to capture the variation, we estimated two versions of the model, one with and one without using the JOLs. We compare the two models on their ability to predict a heldout participant’s data when trained on all of the other participants data (Stone, 1974). One complication here when modeling multiple datasets is that the log-likelihood of the two models is not computed over the same data (since the JOL model also incorporates the likelihood of the judgment). To deal with this, we compare the models only on their ability to predict the recall test trial. Rather than refitting the model several times to hold out each of the subjects individually, for computational efficiency, we use Pareto-Smoothed Importance Sampling, (Vehtari et al., 2017) (as implemented in the `Arviz` package (Kumar et al., 2019)) to approximate the distribution of leave-one-subject-out log likelihoods.

2.2.6 fMRI ANALYSES

The MRI data were analyzed using a two-stage general linear model (GLM) approach. We first estimated the within-participant effects of interest and then computed the group-level across participant stability of the effect for each voxel (Friston et al., 1994; Mumford and Nichols, 2009).

GLM PREDICTORS

Behavioral Predictors We estimated the effect on BOLD data of various predictors derived from individual participants' behavior, from group-level behavioral measures, and from estimated values from our cognitive model. All first-level models included a predictor coding for the timing of study trials (4s boxcar). Predictors based on the raw behavioral responses included: individual participants' Judgments of Learning given to each word pair at the conclusion of the study session (*JOL*); individual participants' recall accuracy after three days for the word pair studied on each trial (*subsequent-memory*); for each word pair, the percentage of participants in the behavioral version of the study who correctly recalled the English word during the memory test at a three day delay (*memorability*).

Model-based Predictors We also used the cognitive model, which incorporates these behavioral data, to estimate a number of latent parameters and these were used to generate model-based predictors. These predictors were obtained by fitting the two-state Markov model described above to all of the behavioral data and then, for each MRI participant, using the forward-backward algorithm (Baum and Eagon, 1967; Baum

et al., 1970), with the posterior over model parameters, to obtain marginal posterior probabilities of being in each state at any point in the protocol, conditioned on that participant’s recall and JOL for that particular word-pair. More formally, we write

$$m_{t,s}^{i,w} = Pr \left[q_t^{w,i} | o_{t_{test},i}^{test,recall}, o_{t_{JOL},i}^{study+JOL,JOL}, l_w, f_w \right] \quad (2.2)$$

where $m_{s,t}^{i,w}$ is the marginal probability that subject i , studying word-pair w was in state s at time t . We use $t_{JOL}^{i,w}$ to indicate the time of the JOL event for subject i studying word-pair w and $t_{test}^{i,w}$ to indicate the timing of the recall test. We can now define the two cognitive variables that we use in the model. For a study trial at time t , $P(s_{Lpost})$, the probability of being in s_L after the study trial is defined as $m_{(t+1),s_L}^{i,w}$. For the same study trial, $P(switch)$ is defined as $m_{t,s_U}^{i,w} l_w$, i.e. the probability of being in s_U before the study trial and the probability of switching given that the word-pair w is presented. This simple definition is possible because a word pair can only switch states if it was previously in the unlearned state s_U . To obtain a single predictor for each variable, we use the posterior mean for each time point.

It may seem strange that we use the probability of a given state as a regressor in the model, which does not directly correspond to the hypothesized state of the brain according to the model in that, if the model were correct, we would not expect any brain region to track the probabilities per se. However, because both variables only have two possible outcomes (s_U or s_L for P_{Lpost} and *switch* or *stay* for $p(switch)$), we can consider the resulting GLM to be fitting a mixture of two time series, one corresponding to each outcome. The probabilities then track the relative weight assigned by the model to each time series and so the inter-

pretation of a brain region tracking a discrete state can still hold. Alternately, as discussed above, our model is a measurement model since many continuous state models (i.e. [Bush and Mosteller, 1953](#)) also result in similar predictions for a probability of recall. In that case, the time series fit by our model may correspond to those continuous states as well.

FIRST LEVEL WITHIN-PARTICIPANT GLMS

Individual participants' preprocessed BOLD data were analyzed using a general linear model (GLM) as implemented in SPM12 ([Friston et al., 1994](#))^{||}. The BOLD activation for each of the five study runs was modeled in a single regression with the design matrix for each model (see below) composed of: predictors coding for the timing and magnitude of potential effects of interest derived from the behavioral task and participant responses; a set of fourteen potential nuisance and confound signals as estimated during preprocessing (the six motion and realignment estimates obtained during preprocessing and an additional eight predictors carrying signal from white matter and CSF ([Behzadi et al., 2007](#))); and five variables coding for the timepoints spanning each of the five separately recorded study runs. The predictors of interest were constructed as four-second boxcars covering the timepoints of their occurrence for each participant. Prior to model estimation these vectors were convolved with the canonical SPM double gamma hemodynamic response function (HRF) along with its' temporal and dispersion derivatives ([Lindquist et al., 2009](#)) to account for the timing of the BOLD signal relative to stimulus events. The nuisance confounds, which were estimated from the data BOLD data, were not convolved with the HRF. The GLM was then fit using ordinary least squares (OLS) resulting in regression coefficients for each

^{||}<https://www.fil.ion.ucl.ac.uk/spm/software/spm12/>

predictor of interest for each voxel summarizing the relationship of these predictors to the voxel timeseries during the course of the study session.

SECOND LEVEL GROUP ANALYSIS

To assess the group-level stability of the effects estimated in the first level within-participant models, we calculated within participant linear contrasts of the estimated regression coefficients relative versus zero resulting in a brain-wide contrast volume for each participant and each comparison of interest. These contrast images were then carried forward to the second level where a one-sample t-test across participants assessed the significance of the contrast at the group level for each voxel. This procedure was repeated for each contrast of interest. The resulting statistical maps were corrected for multiple comparisons using SPMs cluster-level Family Wise Error (FWE) at $p < .05$ (voxel-level cluster forming threshold $p < 0.001$).

2.3 RESULTS

2.3.1 BEHAVIORAL RESULTS

Figure 2.3 shows key features of the behavioral data. Across the three different test delays, memory performance generally drops, likely due to forgetting, as would generally be expected in a memory experiment. Participant overall performance varied widely from 0 to 100 percent correct. We test for differences between recall performance between the *fMRI* participants and the *Behavior-only* participants at the same delay using a generalized linear mixed effect model (as implemented in lmer (Bates et al., 2015)) with random effects for word pair and subject. We find no significant differences in performance between the two groups ($p = .6729$). In addition, perhaps surprisingly, as seen in Figure 2.3, participants'

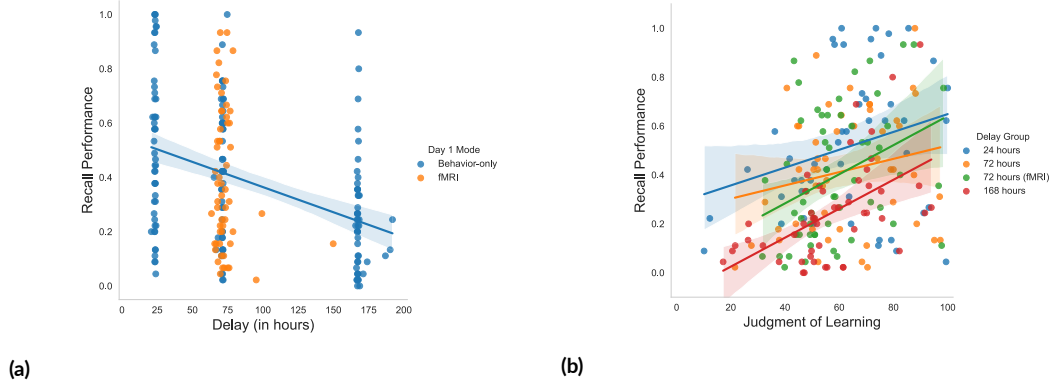


Figure 2.3: 2.3a: Mean recall performance (% correct) for individuals (dots) at each study-test delay. 2.3b: Mean individual participant Judgment of Learning is correlated with individual overall percent recalled within each delay condition.

JOLs were reasonably calibrated across-subjects as a participant’s overall performance was moderately correlated with their average JOLs following study ($r(199) = .35, p < 1e-6$).

2.3.2 MODEL RESULTS

First to check that our model was able to capture the variation in the data, we conducted a posterior predictive check (Gelman et al., 1996, 2000), in which we took each set of samples of l_w and f_w and generated correct or incorrect recalls for each subject based on the model, their timing and their JOL response (i.e., using $\pi_{test}^{w,i,s'}$ as described in the Methods section). This gave us 300 predictive datasets which we plot in Figure 2.4 alongside the true dataset for 9 arbitrarily selected word-pairs. We can see that the model is largely able to capture the variation across the words with most of the realized data falling within 1 standard deviation of the mean predictive dataset.

We also wanted to test whether our use of the JOLs to fit the model actually improved our ability to capture the data. To do so, we fit two versions of the model, with and without the judgments of learning. We then used a computationally efficient approximation

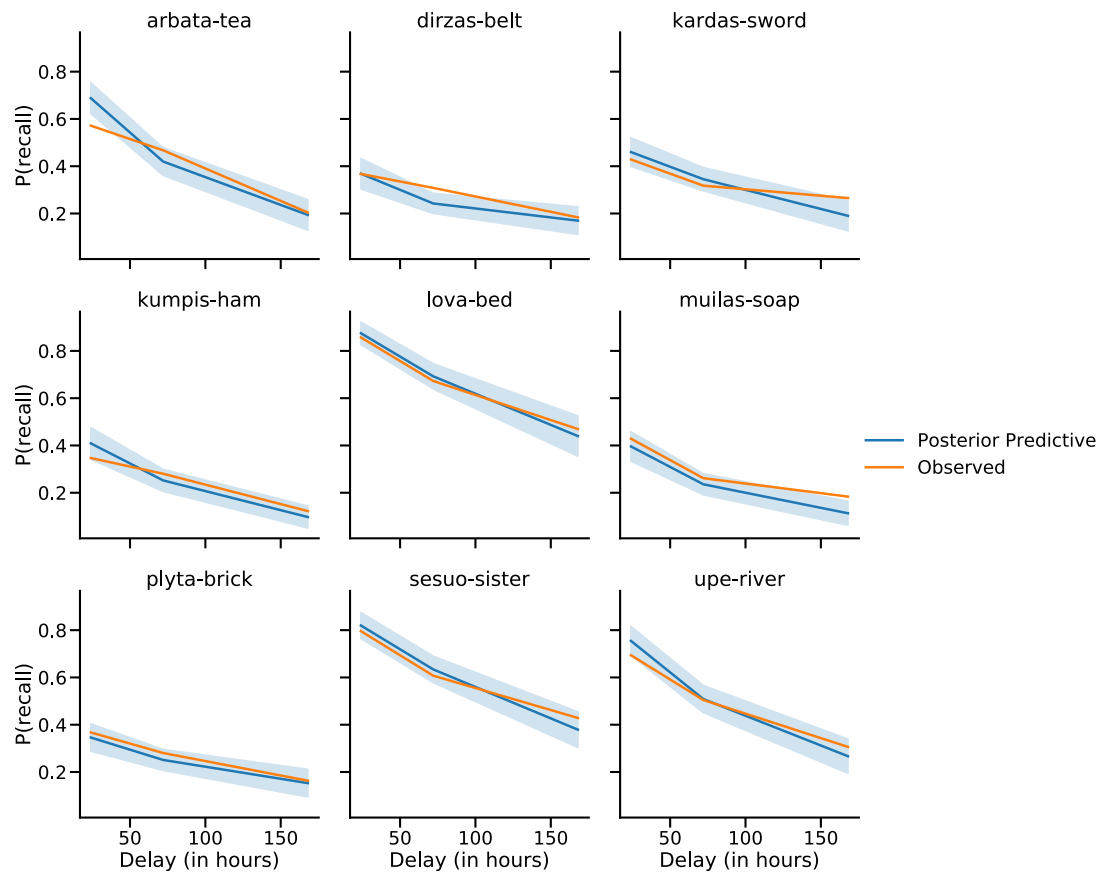


Figure 2.4: Posterior predictive check for 9 of the 45 Lithuanian-English word-pairs, selected arbitrarily by choosing every 5th one when sorted in Lithuanian alphabetical order. Plots show the true proportion correct in the data in orange. 300 posterior predictive datasets were generated by based on sampled parameters and the average recall accuracy was computed for each word in each one. Blue lines show the mean across these datasets and shaded areas show ± 1 SD

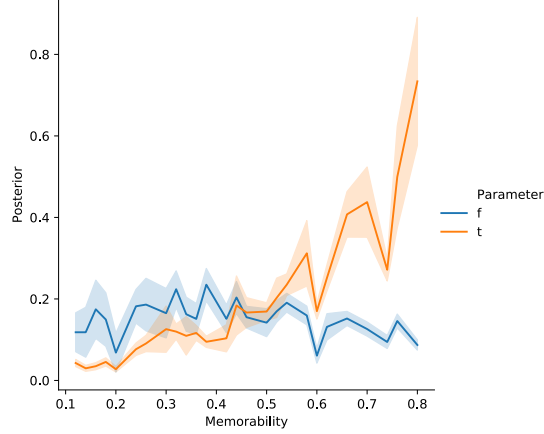


Figure 2.5: Posterior mean ± 1 SD for f_w and t_w parameters as a function of the word-pair memorability at a three-day delay

(Vehetari et al., 2017) the distribution of leave-one-subject-out (LOSO) log likelihoods for each model. We found that the model with JOLs had a predicted log likelihood that was over three standard errors greater than the model without (mean approximate LOSO log likelihood difference = 296.6, $SE = 98.5$), suggesting that it better accounted for the data even of subjects not used to fit the model.

As asserted above, learnability and forgetability are essential to predicting the latent memory dynamics. We first demonstrate that the learnability and forgetability parameters inferred by our model are not necessarily correlated with each other (as suggested by Woodworth, 1914) or with more standard measures of average memorability at the three-day delay (Bainbridge, 2019). Figure 2.5 shows the posterior over the f_w and t_w model parameters as a function of the average memorability after a 3 day delay. We can see that these measures are neither perfect correlated with each other or with memorability. For instance, a word-pair can have a higher memorability than another when it has both lower learnability and lower forgetability.

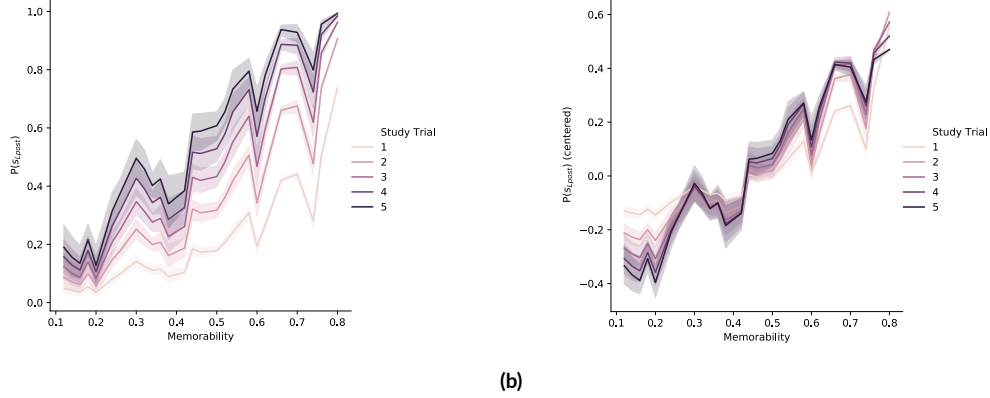
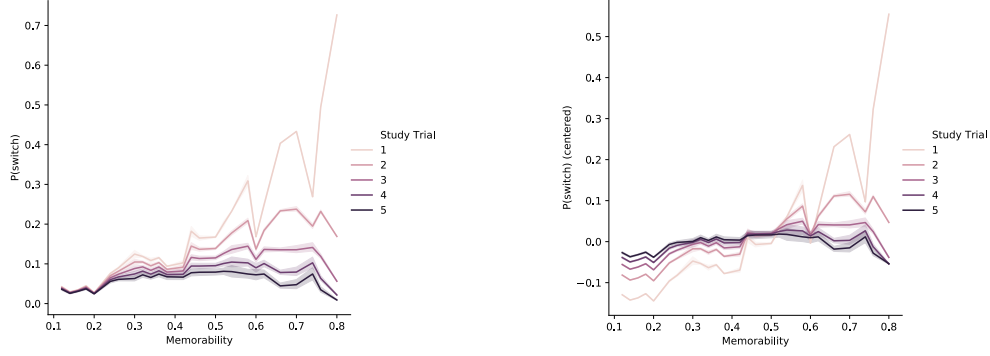


Figure 2.6: 2.6a: Posterior means for $P(s_{Lpost})$ averaged over participants for each study trial and word-pair as a function of that word-pair's memorability. 2.6b: Same plot but with $P(s_{Lpost})$ centered at 0 to emphasize how predictions vary across runs. Confidence intervals represent bootstrapped standard errors.

We now want to understand how our two cognitive variables predict the fMRI data and how it makes trial by trial predictions over runs. In general, we plot these predictions against memorability, another cognitive variable that has been shown to predict significant activity during memory tasks (Bainbridge et al., 2017; Bainbridge and Rissman, 2018) to emphasize the novel part of the variance explained by our model-based variables. In Figure 2.6, we plot the posterior mean for $P(s_{Lpost})$ for each word-pair, averaged over fMRI subjects, as a function of memorability and study presentation. This plot shows how this variable increases overall across trials but at different rates for different words, in a way that depends on the learnability. The variation across subjects is due to both variation in the subject's performance and judgments, as well as slight variation in the timing across runs and final test times. Ultimately, because the first-level GLM analysis is only within subject and run, we also plot a version of this variable that is centered around 0 for each study presentation. This emphasizes the variation that is captured by this parameter and how it changes across runs. Figure 2.7 plots the same thing for the $P(switch)$ parameter.



(a)

(b)

Figure 2.7: 2.7a: Posterior means for $P(\text{switch})$ averaged over participants for each study trial and word-pair as a function of that word-pair’s memorability. 2.7b: Same plot but with $P(\text{switch})$ centered at 0 to emphasize how predictions vary across runs. Confidence intervals represent bootstrapped standard errors.

2.3.3 MRI RESULTS

To characterize the relationship between brain activation during the study session and observed as well as latent memory dynamics we estimated several separate GLMs. We began by estimating a classic subsequent memory GLM in which study trial activity was modeled according to binary recall accuracy on later memory test (section 2.3.4). Then, to assess whether latent memory-related quantities in our cognitive model were related to brain activation we estimated GLMs using the $p(\text{switch})$ and $p(s_{Lpost})$ values obtained by evaluating the fitted cognitive model conditioned on each MRI participants’ JOLs and recall accuracy. This gives us a set of posterior mean probabilities for each person on each study trial. These values were then included as predictors in first-level within-participant GLMs (section 2.3.5).

2.3.4 SUBSEQUENT MEMORY GLM

We first report analyses of the study session brain data using a conventional subsequent memory approach. In this model the activity on each study trial is modeled according to whether that participant later responded correctly (*Rem*) or incorrectly (*Forg*) on the cued recall test. For a given word pair and a particular participant this gives the same value for each of the five study trials. Multiple-comparison thresholded results of this analysis are shown in Figure 2.8. Significant clusters showing greater study trial activation for later remembered compared to later forgotten word pairs were observed in bilateral parietal cortex, bilateral lateral occipital regions including the putative visual word form area (VWFA) (Cohen et al., 2002; Chen et al., 2019), left prefrontal cortex (PFC) and bilateral cerebellum. There was an additional cluster in the posterior part of the right superior temporal gyrus (MNI: 60, -22, 13) that showed greater activation for later forgotten word pairs. These regions are generally consistent with prior subsequent memory studies, overlapping substantially with clusters reported in a metaanalysis of univariate subsequent memory studies Kim (2011).

2.3.5 MODEL-BASED GLMS

The subsequent memory GLM results showed activations consistent with roles of parietal, prefrontal, and lateral occipital areas in successful memory encoding. However, this model almost certainly doesn't adequately capture a learner's trajectory through our task because the GLM predictors, based on later recall, assume a constant level of activation on each study trial for a particular word. The subsequent memory model additionally reflects memory only as tested at a three day delay which doesn't capture whether a particular word

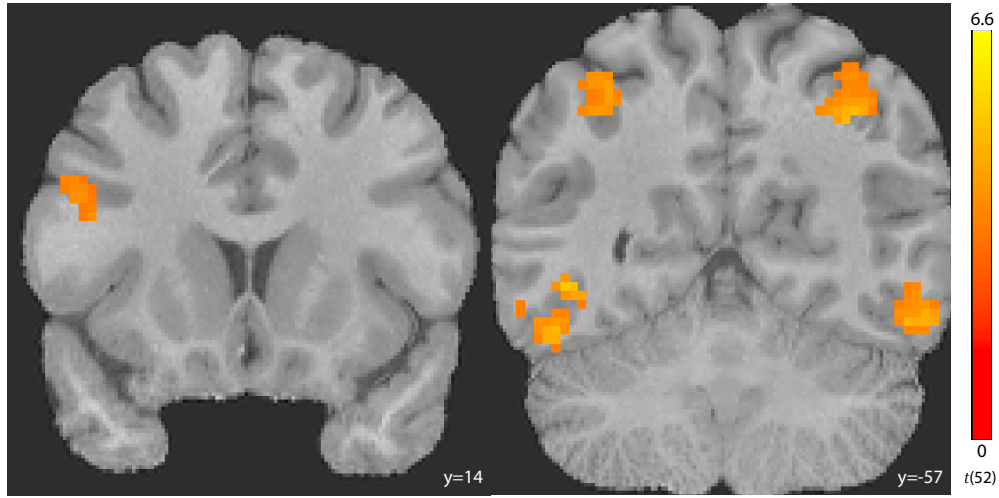


Figure 2.8: Thresholded t-stat maps showing clusters with significantly greater activation on study trials for later remembered compared to forgotten word pairs. Left panel: left prefrontal cortex (MNI: -44, 13, 22); Right panel: bilateral parietal and lateral occipital cortical clusters.

pair might have been learned at the time of study and later forgotten. One strength of our cognitive model is that it can provide estimates of the dynamics of memory acquisition during the course of the repeated study trials. To assess whether these estimated dynamics were reflective of brain activation we fit two additional GLMs using the probability that a particular word-pair was in memory at the end of each study trial $p(s_{Lpost})$ and the probability that a word-pair switched states from unlearned to learned on each study trial $p(switch)$.

GLMs USING PROBABILITY OF LEARNED STATE

Example results of the $p(s_{Lpost})$ model are shown in Figure 2.9. This model identified robust activation throughout the brain where clusters of voxels tracked the probability of an item being known at the time of each study trial. The clusters identified in this model after FWE correction overlapped substantially with the subsequent memory clusters but included additional regions. Sixty-six percent of the above-threshold subsequent mem-

ory voxels identified in the previously reported GLM overlapped with the $p(s_{Lpost})$ model, and with the exception of the negative subsequent memory effect in superior temporal gyrus the non-overlapping subsequent memory voxels were all immediately adjacent to the $p(s_{Lpost})$ regions, as shown in the green outlines of Figure 2.9. We observed additional clusters associated with the $p(s_{Lpost})$ model in bilateral cerebellum, along the left middle temporal gyrus and fusiform gyrus, bilateral parietal cortex, along the dorsal portion of anterior left hippocampus stretching into the amygdala and the tail of the caudate, left caudate body, extensive dorsal and ventral left prefrontal regions including anterior insula regions associated with memory encoding Kim (2011), and dorsomedial prefrontal cortex. Clusters negatively associated with this predictor were primarily located in posterior portions of the visual cortex.

GLMs USING PROBABILITY OF SWITCHING ON STUDY TRIALS

Results from the $p(switch)$ model are shown in Figure 2.10. This model identified regions positively associated with the probability of switching memory states on each study trial in right lateral cerebellum (crus II), left lateral parietal cortex, posterior portions of the right superior temporal sulcus, left anterior middle temporal gyrus, and left inferior frontal cortex. As in the $p(s_{Lpost})$ GLM there was significant negative association with this predictor in posterior portions of the visual cortex.

MODEL-BASED PREDICTORS AND BEHAVIORAL MEASURES

The above analyses show that model based predictors tracking learning dynamics over time can account for study session brain activation in a number of regions, including those iden-

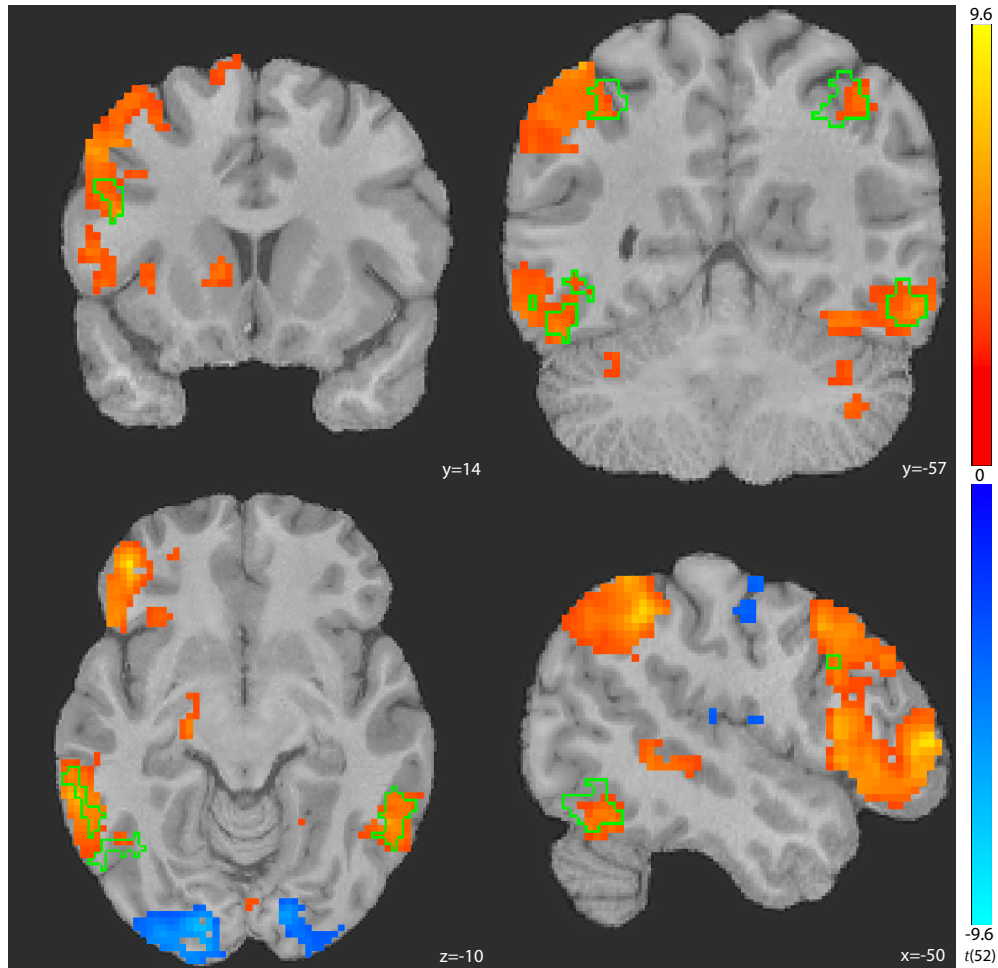


Figure 2.9: Thresholded t-stat maps showing clusters significantly associated with the probability of a word pair being in the *Learned* state on each trial. Green outlines show the location of significant subsequent memory clusters as reported in section 2.3.4

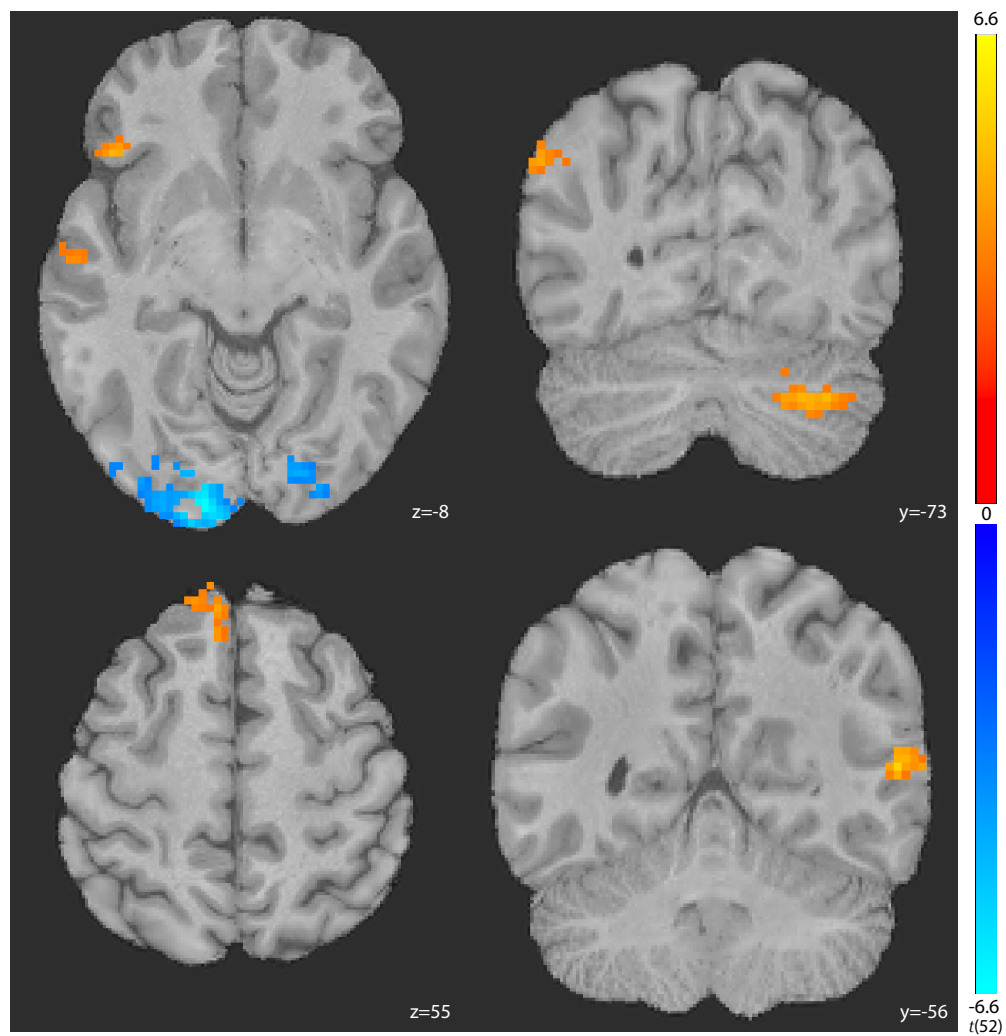


Figure 2.10: Thresholded maps showing regions associated with the *probability of switching* memory states on each trial.

tified in a subsequent memory analysis. However, the model-based predictions are derived from a combination of individual JOLs, individual recall performance, and group level performance (*memorability*) on each word pair. To explore whether these single model-based predictors were accounting for brain activation above that tracked by the raw JOL, memorability, and recall accuracy measures, we conducted additional analyses in which we modeled in the brain data in a GLM including these three behavioral measures alongside the model based $p(s_{Lpost})$ or $p(switch)$ predictors (Figure 2.11). If we do find brain regions where the model predicts beyond these measures, this suggests that there is some additional variation that the model accounts for, either by nonlinearly combining the input data or in how the model predictors vary over trials, as shown in Figure 2.6 and 2.7. This is therefore a very strong test of the model’s value in structuring the neuroimaging analysis.

PROBABILITY OF LEARNED STATE AND RAW BEHAVIORAL DATA In the GLM including the behavioral measures with the $p(s_{Lpost})$ predictor there was still a significant left lateral parietal cortex in the vicinity of the angular gyrus that was associated with the probability of being in the known state (centered at MNI -38, -64, -37; figure 2.11A). This cluster was a subset of the voxels identified in the GLM using the $p(s_{Lpost})$ on its’ own. Recent work has shown that this region tracks the contents of memory retrieval (Kuhl and Chun, 2014; Xiao et al., 2017; Favila et al., 2018) and that the degree of univariate activation is correlated with the vividness of memory recollection (Lee et al., 2019; Daselaar, 2009). This suggests that in our data, a higher probability of an item being learned after a particular study trial may have been associated with successfully retrieving previously formed memories for that word-pair.

A subset of the visual cortical ROIs that were negatively associated with $p(s_{Lpost})$ in the

single predictor model remained above threshold in this model. Interestingly there were also some new clusters in dorsal parietal cortex identified in this model, and a number of the active clusters in the single predictor model were not identified in this fuller model, but this is not surprising given that the cognitive model from which the $p(s_{Lpost})$ values were estimated is based on the additional predictors included in these GLMs. To the extent that the model-based predictors are in some sense combinations of those behavioral measures it is noteworthy that the model-based predictors identified significant activations above those associated with the raw behavioral measures. These results support the cognitive model as leveraging the behavioral measures and the model structure to provide additional information about the dynamics of memory states during the course of learning that can be used to understand brain activation.

PROBABILITY OF SWITCHING AND BEHAVIORAL DATA The GLM including the model-based $p(switch)$ predictor alongside the JOLs, memorability, and individual participant recall accuracy measures also identified significant clusters associated with the model-based predictor. An above threshold cerebellar region positively associated with $p(switch)$ in the simple model remained in this fuller model, as did the posterior visual cortical regions that had previously been identified as negatively related to the $p(switch)$ estimates. The cerebellar region remaining in this model (MNI 30, -78, -37) is intriguing in that it has previously been reported to be associated with encoding related signals (Maillet and Rajah, 2014) as well as tasks focused on semantic processing and reading (McDermott et al., 2003; Alvarez and Fiez, 2018) as might be expected of an area contributing to the learning of a foreign language vocabulary.

More broadly and beyond interpretation of individual ROIs, we believe the the robust-

ness of the GLMs using the model-based predictors alone, including their overlap with activations associated with the raw behavioral measures, is of particular interest for future work understanding the dynamics of learning over time and suggests that such measures could fruitfully be employed in cases where one does not have access to an individual learner’s JOLs or later recall accuracy.

2.4 GENERAL DISCUSSION

In the current work, we leveraged a well-established memory model to track mnemonic status of individual items in a language-learning paired-associates recall task. This cognitive model allowed us to operationalize two types of variables related to mnemonic status that might be reflected in neural signals, namely, the probability that an item is learned after the current trial ($P(s_{L_{post}})$) or the probability that the current trial is the one in which learning occurred ($P(switch)$).

Using a large dataset, we showed that these two variables, both of which varied over trials depending on the word-pair being studied and the participants’ own recall performance and confidence judgments, could predict fMRI BOLD activity in several brain regions, with different regions being correlated with different variables. Notably, these included many regions that are typically observed in studies of memory but, due to limitations of the methods used to study them, are typically only found in studies with single learning trials. Relative to using straightforward applications of traditional techniques in the cognitive neuroscience of memory, a much wider subset of voxels was shown to be involved in tracking the mnemonic status using the variables derived from our cognitive model. In addition, by adjusting for the various inputs into the cognitive model, we were

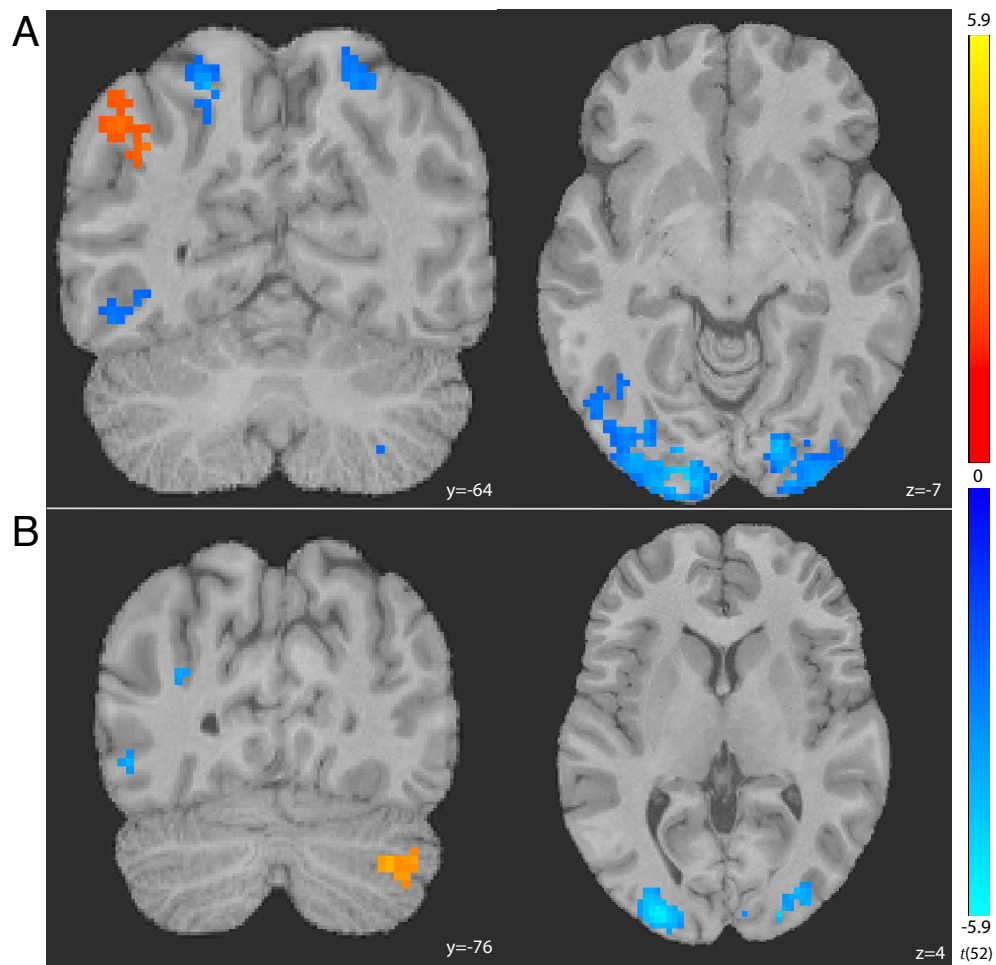


Figure 2.11: Results from GLMs including raw behavioral measures alongside model-based predictors. A) Coronal (left) and axial (right) slices showing significant clusters associated with $p(s_{Lpost})$; B) significant clusters associated with $p(switch)$

able to demonstrate that the structure of the model itself, and how it combined the data nonlinearly and over study presentations, were key in making these predictions in at least some of the regions. These regions included a parietal ROI where increased activity has often been associated with repeated presentations and stimulus familiarity (Gilmore et al., 2019b,a) and strong, vivid recollection (Lee et al., 2019; Daselaar, 2009). The relationship between $P(s_{L_{post}})$ and the parietal lobe is consistent with accounts of the fidelity of reinstatement contributing to the improvement of memory with practice (Benjamin and Tullis, 2010; Xue, 2018; Xue et al., 2013). We hope that further work with this model can help elaborate these mechanisms in a more quantitative way. While we only investigate univariate activity here, future work may be able to use this model to investigate multivariate activity which is often used to support claims of reactivation during study (Xue et al., 2010; Ward et al., 2013).

While some of the variance explained by the model may be accounted for by simpler behavioral measures, the model-based measures on their own isolated several other regions often thought to be key for learning and memory, including a caudate region similar to one that Berens et al. (2018) found in a task where they also found support for a related all-or-none learning model using a model-based analysis. In addition, from a parsimony perspective, the amount of variation accounted for by the model variables is impressive as it uses only a single variable rather than the three behavioral variables we adjusted for in the final GLM models.

A final key feature of the model is the ability to identify two potential contributions to a subsequent memory effect, the probability of switching memory states on a study trial and the probability that an item was learned after study. While these two are confounded

in a single trial, the model revealed that, over multiple trials, these variables become uncorrelated. In this analysis, we find separate regions of the brain that are related to each of these variables suggesting that these two variables underlie different computations in the brain. This suggests that this may be a fruitful approach to understanding the dynamics of memory over time and their neural correlates.

While the overall success of this model is promising, it is still limited in many ways and could be improved. Many parameters were difficult to identify and were fixed to values based on past work with similar models. There may be better choices for these values and, indeed, better, more complex model structures that account for the true dynamics of learning and memory. In this paper, we showed one approach to identifying parameters by collecting a cheaper behavioral dataset. Future work could potentially collect data in more conditions, following a strategy we develop in more detail in chapter 3. In addition, one simple addition to the model would be allowing the learning and forgetting parameters to vary by subject, constrained by a hierarchical model. While we started with a simple version for a proof of concept, this would be a promising future direction that would likely help the model identify more relevant activity.

One methodological detail of this work is the use of the analytic posterior for the Markov model states given the observed data that allowed us to individualize the state predictions even when the model parameters were fit at the group level. This is an approach that we believe may be particularly fruitful for model-based fMRI analyses as trial-by-trial predictions can be even more tightly correlated with the specific subject's behavior, even when that behavior is temporally distant from the neural signal (as it was in our task). As we explore in chapter 3, this correlation is key to increasing power in model-based analyses.

In particular, the approach we laid out also has the potential to easily incorporate more diverse types of data such as reaction times and eye movements, which may improve predictions even more and we suggest that future work attempt this. Some readers may even question why we did not build the fMRI signals into the model itself as we did with the behavioral signals, an approach that is known as joint modeling (Turner et al., 2013, 2019). While perhaps more statistically efficient and, in some ways, more aesthetically pleasing (to us), this approach is quite computationally expensive, especially if we do not know in advance what signals would be relevant. However, we fully believe that this is ultimately the preferred approach and expect that future work will develop in this direction.

Overall, the results obtained in this study appear to hold promise for the use of model-based fMRI in studies of memory (Love, 2020), a topic which is ideally suited for the use of models as it's targets are often not directly related to observable measures and it unfolds over long periods of time. We hope that this study is only the beginning of exploring the complex dynamics of learning and memory over time and their representation in the brain.